Extra Dessert #15: Miscellaneous Rules

External RF amplifiers, prohibited communications, spread spectrum, auxiliary stations, Canadian reciprocity, Line A, STA.

Group E1F · 11 pool questions · Self-study reference · Practice exams at eham.net/exams

Spread Spectrum

Spread spectrum is permitted only on amateur frequencies above 222 MHz (E1F01).

External RF Power Amplifiers

Question	Answer	Q
When may a dealer sell an external RF power amp (operating below 144 MHz) without FCC certification?	**When the amplifier is constructed or modified by an amateur radio operator for use at an amateur station** (E1F03). Hams get a homebrew exception.	E1F03
Standard that must be met for FCC certification of an external RF amp	**It must satisfy the FCC's spurious emission standards when operated at the lesser of 1500 watts or its full output power** (E1F11).	E1F11

Line A (US-Canada Border)

Question	Answer	Q
Geographic description of "Line A"	**A line roughly parallel to and south of the border between the US and Canada** (E1F04).	E1F04
Frequency segment NOT permitted in the contiguous 48 states north of Line A	**420 MHz - 430 MHz** (E1F05). Protects Canadian government radiolocation in that segment.	E1F05

Canadian Operators In The US

What privileges are authorized in the US for persons holding a Canadian amateur license? The operating terms and conditions of the Canadian amateur service license, not to exceed US Amateur Extra class license privileges (E1F02). They keep their home privileges, capped at US Extra.

Prohibited Communications

Question	Answer	Q
When may an amateur station send a message to a business?	**When neither the amateur nor their employer has a pecuniary interest in the communications** (E1F07). No money on the line.	E1F07
Type of amateur communications that are prohibited	**Communications transmitted for hire or material compensation, except as otherwise provided in the rules** (E1F08).	E1F08
What cannot be transmitted over a mesh network?	**Messages encoded to obscure their meaning** (E1F09). The general prohibition on obscured/coded messages still applies.	E1F09

Auxiliary Stations And STA

Question	Answer	Q
Who may be the control op of an auxiliary station?	**Only Technician, General, Advanced, or Amateur Extra class operators** (E1F10). Novice excluded.	E1F10
When might the FCC issue a Special Temporary Authority (STA)?	**To provide for experimental amateur communications** (E1F06).	E1F06

Dessert 15 Question Bank (E1F)

E1F01. On what frequencies are spread spectrum transmissions permitted?
A. Only on amateur frequencies above 50 MHz.
B. Only on amateur frequencies above 222 MHz.
C. Only on amateur frequencies above 420 MHz.
D. Only on amateur frequencies above 144 MHz.

E1F02. What privileges are authorized in the US to persons holding an amateur service license granted by the government of Canada?
A. None, they must obtain a US license.
B. Full privileges of the General class license on the 80-, 40-, 20-, 15-, and 10-meter bands.
C. The operating terms and conditions of the Canadian amateur service license, not to exceed US Amateur Extra class license privileges.
D. Full privileges, up to and including those of the Amateur Extra class license, on the 80-, 40-, 20-, 15-, and 10-meter bands.

E1F03. Under what circumstances may a dealer sell an external RF power amplifier capable of operation below 144 MHz if it has not been granted FCC certification?
A. Gain is less than 23 dB when driven by power of 10 watts or less.
B. The equipment dealer assembled it from a kit.
C. It was manufactured and certificated in a country which has a reciprocal certification agreement with the FCC.
D. The amplifier is constructed or modified by an amateur radio operator for use at an amateur station.

E1F04. Which of the following geographic descriptions approximately describes “Line A”?
A. A line roughly parallel to and south of the border between the US and Canada.
B. A line roughly parallel to and west of the US Atlantic coastline.
C. A line roughly parallel to and north of the border between the US and Mexico.
D. A line roughly parallel to and east of the US Pacific coastline.

E1F05. Amateur stations may not transmit in which of the following frequency segments if they are located in the contiguous 48 states and north of Line A?
A. 440 MHz - 450 MHz.
B. 53 MHz - 54 MHz.
C. 222 MHz - 223 MHz.
D. 420 MHz - 430 MHz.

E1F06. Under what circumstances might the FCC issue a Special Temporary Authority (STA) to an amateur station?
A. To provide for experimental amateur communications.
B. To allow use of a special event call sign.
C. To allow a VE group with less than three VEs to administer examinations in a remote, sparsely populated area.
D. To allow a licensee who has passed an upgrade exam to operate with upgraded privileges while waiting for posting on the FCC database.

E1F07. When may an amateur station send a message to a business?
A. When the pecuniary interest of the amateur or his or her employer is less than $25.
B. When the pecuniary interest of the amateur or his or her employer is less than $50.
C. At no time.
D. When neither the amateur nor their employer has a pecuniary interest in the communications.

E1F08. Which of the following types of amateur station communications are prohibited?
A. Communications transmitted for hire or material compensation, except as otherwise provided in the rules.
B. Communications that have political content, except as allowed by the Fairness Doctrine.
C. Communications that have religious content.
D. Communications in a language other than English.

E1F09. Which of the following cannot be transmitted over an amateur radio mesh network?
A. Third party traffic.
B. Email.
C. Messages encoded to obscure their meaning.
D. All these choices are correct.

E1F10. Who may be the control operator of an auxiliary station?
A. Any licensed amateur operator.
B. Only Technician, General, Advanced, or Amateur Extra class operators.
C. Only General, Advanced, or Amateur Extra class operators.
D. Only Amateur Extra class operators.

E1F11. Which of the following best describes one of the standards that must be met by an external RF power amplifier if it is to qualify for a grant of FCC certification?
A. It must produce full legal output when driven by not more than 5 watts of mean RF input power.
B. It must have received an Underwriters Laboratory certification for electrical safety as well as having met IEEE standard 14.101(B).
C. It must exhibit a gain of less than 23 dB when driven by 10 watts or less.
D. It must satisfy the FCC’s spurious emission standards when operated at the lesser of 1500 watts or its full output power.

Answer Key

E1F01: B	E1F02: C	E1F03: D	E1F04: A
E1F05: D	E1F06: A	E1F07: D	E1F08: A
E1F09: C	E1F10: B	E1F11: D	